

Piyush Paul

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piyushpaul.com — West Bengal, India

Experience

Backend Developer Intern,

Fluencer Market (Remote)

Dec 2024 – Apr 2025

- Designed and developed RESTful APIs using Flask to support influencer-brand collaboration workflows.
- Optimized backend performance, reducing server latency by 35% through efficient query design and data modeling.
- Collaborated with cross-functional teams to deliver scalable and production-ready backend services.

Projects

ARTA — Autonomous Red Team Agent

Apr 2026 – Present

FastAPI, Next.js, Gemini AI, SQLite, Puppeteer, Nmap, Metasploit

- Built an end-to-end autonomous penetration testing pipeline (Recon → Vulnerability Analysis → Exploit → Report) using AI agents.
- Integrated real-time CVE detection via NVD API and automated exploit validation with sandboxed execution.
- Developed a real-time dashboard using SSE in Next.js for live scan tracking, logs streaming, and findings visualization.
- Automated PDF report generation with structured vulnerability insights, impact analysis, and remediation steps.

AstroRoni

Sep 2025 – Apr 2026

Next.js, Supabase Auth, Flask, MongoDB, Razorpay, OpenAI, GCP

- Developed a full-stack astrology SaaS platform generating Vedic Kundli and numerology reports using AI.
- Integrated Razorpay for payments and Supabase Auth for secure authentication and subscription handling.
- Designed a scalable microservices backend deployed on Google Cloud Platform.

RiskRadar

Nov 2024 – Apr 2025

Next.js, Kinde Auth, MongoDB, Stripe, Gemini AI, GCP

- Built a vulnerability tracking system that monitors CVEs in real time and sends targeted alerts.
- Used Gemini AI to generate prioritized remediation strategies, reducing response time for users.
- Implemented secure authentication and subscription billing using Kinde Auth and Stripe.
- Deployed on GCP ensuring scalability and high availability.

Technical Skills

Languages: JavaScript, TypeScript, Python, C++, Java, Rust, SQL, HTML, CSS

Frontend: React.js, Next.js, Tailwind CSS

Backend & Databases: Node.js, Flask, Spring Boot, Strapi, Supabase, MongoDB, SQL

Cloud & DevOps: Google Cloud Platform (GCP), Docker, Git

AI/ML & APIs: OpenAI API, Gemini AI, RESTful API Design

Tools: IntelliJ IDEA, Postman, Microsoft Office Suite

Achievements

MLH Hackathon — 1st Runner-Up

2026

- Secured 1st Runner-Up position at a Major League Hacking (MLH) hackathon by building **NeighbourNet**, a mesh networking solution, demonstrating strong problem-solving, rapid prototyping, and pitching skills under tight deadlines.

Winner — MLH Hackdays

2026

- Won MLH Hackdays by showcasing **ARTA (Autonomous Red Team Agent)**, an AI-powered penetration testing system.

Frontend Hackathon — 2nd Place, West Bengal, India

2024

- Secured 2nd position among thousands of participants by demonstrating strong UI/UX and rapid prototyping skills.

Education

JIS College of Engineering, West Bengal, India

Bachelor of Technology in Computer Science

Jun 2025 – Present

- CGPA: 7.9/10
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Database Management Systems